

■ A.6 Graphics Programming

■ A.6.1 Pictures of Polyhedra

As well as storing various numerical properties of polyhedra, you can also use *Mathematica* definitions to specify the geometrical properties of polyhedra, in a form that can be converted to pictures using *Mathematica* graphics functions.

```
Icosahedron::usage =
  "Icosahedron[{pos:{0,0,0}, (scale:1)}] gives a graphics
  object representing an icosahedron with specified center
  position and scale."

MakePolyhedron[coords_, pos:{_?NumberQ, _?NumberQ, _?NumberQ}, scale_?NumberQ] :=
  Map[Polygon, Map[# + pos &, scale coords, {2}] ]
MakePolyhedron[coords_, pos_] := MakePolyhedron[coords, pos, 1.]
MakePolyhedron[coords_] := MakePolyhedron[coords, {0.,0.,0.}, 1.]

Icosahedron[opts___] := MakePolyhedron[IcosahedronCoords, opts]

Stellate::usage =
  "Stellate[polyhedron, (stellation ratio:2)] generates a stellated
  polyhedron from a polygon list. Negative stellation ratios give
  concave figures."

StellateFace[face_List, k_] :=
  Block[ { apex,
    n = Length[face] } ,
    apex = N [ k Apply[Plus, face] / n ] ;
    Table[ Polygon[ {apex, face[[i]], face[[ Mod[i, n] + 1 ]] }
      ],
      {i, n} ]
  ]

Stellate[poly_, k_:2?NumberQ] :=
  Flatten [ poly /. Polygon[x_] :> StellateFace[x, k] ]
```

Geometrical structure of polyhedra.

Read in the file of polyhedra.

```
In[1]:= <<Polyhedra.m
Out[1]= Polyhedra`
```

This generates a graphical object that represents an icosahedron.

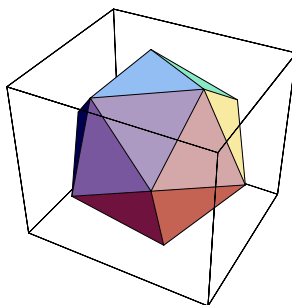
```
In[2]:= Icosahedron[ ] // Short
Out[2]//Short= {Polygon[<<1>>], <<18>>, Polygon[{<<3>>}]}
```

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This makes a picture of the icosahedron.

```
In[3]:= Show[Graphics3D[Icosahedron[ ]]]
```



Stellate takes the symbolic representation of the icosahedron, and produces a representation of a stellated icosahedron.

```
In[4]:= Show[Graphics3D[Stellate[Icosahedron[ ]]]]
```

